KCM Heat-Shock Transformation

`transformation`

BLOCKS

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block A — 4 °C
block B — 42 °C
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Turn on the **EchoTherm** and wait until both blocks are at temperature. Keep them there throughout.

Alternatives: a thermocycler with two blocks, or a 42 °C heating block and an ice bath.

CELLS

1. Put a **100 µL** competent cell aliquot on block A and thaw (~30 s).
2. Add **25 µL KCM** and pipette gently to mix. Keep it on block A.

25 µL into a 100 µL aliquot is **1× KCM** from the 5× stock. One aliquot does about **3 reactions**.

TRANSFORM

1. Put the DNA tube on block A and let it cool for 30 s.
2. Add **40 µL** of the cell/KCM mix to the DNA. Mix gently.
3. Hold at **4 °C for 10 min**.
4. Move to block B — **42 °C for 90 s**.
5. Back to block A — **4 °C for 1 min**.

HOW MUCH DNA

- **40 µL cells to 10 µL DNA** is the standard. It assumes the DNA is about 20% of the total.
- A large reaction (~20 µL): use 100 µL, or the whole tube of cells.
- Retransforming a miniprep: **0.5 µL** plasmid into 10 µL cells is plenty.
- Pure plasmid: dilute 10–20× in water first and use 1 µL of that.
- Less DNA is fine. Too much dilutes the salts and **drops the efficiency**.

PLATE

1. Plate everything on **Carb**. Incubate **inverted**, 37 °C, overnight.
2. Cancel the temperature programs when you are done.

> **No rescue needed.** Amp/Bla selection uses the carbenicillin plates in the fridge, and plates directly.

LABELLING

Warm plates to room temperature first — cold plates are hard to write on.

On the **bottom**: date, your initials, strain **Mach1**, plasmid **plasmid_name**, selection **Carb**.

Plating more than 100 µL: leave the plate uncovered at the sterile bench until the surface is no longer glossy, or the colonies bleed.